🐼 Pandas EDA

Cheat Sheet for Data Analysts

Essential functions every data professional must know

Swipe to explore →

📥 Data Loading & Quick Inspection

Load Your Data

```
df = pd.read_csv('file.csv')
df = pd.read_excel('file.xlsx')
df = pd.read_sql(query, connection)
```

Understand Structure

```
df.info()        # data types + nulls
df.describe()    # stats summary
df.dtypes        # column types
```

Quick Peek

```
df.head()       # top rows
df.tail()       # bottom rows
df.shape        # (rows, cols)
```

Check Missing Values

```
df.isnull().sum()         # count nulls
df.fillna(value)          # fill nulls
df.dropna()               # drop nulls
```

💡 Pro Tip: Always call df.info() and df.describe() first — they reveal shape, dtypes, and missing data in one shot.

🧹 Data Cleaning & Transformation

Rename & Drop Columns

```
df.rename(columns={'old':'new'})
df.drop(columns=['col_name'])
```

Duplicates

```
df.duplicated()         # find dupes
df.drop_duplicates()    # remove them
```

Type Conversion

```
df['col'].astype('category')
pd.to_datetime(df['date_col'])
```

Apply & Transform

```
df['col'].apply(lambda x: x + 1)
df.apply(lambda row: func(row), axis=1)
```

Pivot & Reshape

```
df.pivot_table(index='col1',
  values='col2', aggfunc='mean')
df.melt(id_vars=['col1'])
```

Merge DataFrames

```
pd.merge(df1, df2, on='col', how='left')
pd.concat([df1, df2])
```

🔍 Smart Indexing & Selection

Select by Position

```
df.iloc[0:5]        # rows
df.iloc[:, 0:3]    # cols
```

Select by Label

```
df.loc[0:5]
df.loc[:, 'col1':'col3']
```

Conditional Filter

```
df[df['col'] > 10]
df[df['col'].isin([1,2,3])]
```

Query Function (SQL-style)

```
df.query('age > 25 and salary > 50000')
```

Statistical Analysis

```
df.corr()                # correlation matrix
df['col'].value_counts()   # frequency count
df['col'].unique()         # distinct values
```

📊 GroupBy & Aggregation

Basic GroupBy

```
df.groupby('col').mean()
df.groupby(['col1','col2']).sum()
```

Multiple Aggregations

```
df.groupby('col').agg(
  ['mean','sum','count'])
df.groupby('col').agg(
  {'a':'mean','b':'sum'})
```

Time Series Grouping

```
# Group by month
df.groupby(pd.Grouper(
  key='date_col', freq='M')).sum()
```

Resample & Rolling

```
df.resample('M', on='date').mean()
df['col'].rolling(window=5).mean()
df['col'].shift(1)   # lag analysis
```

Normalize & Standardize

```
# Min-Max:  (df['col'] - df['col'].min()) / (df['col'].max() - df['col'].min())
# Z-Score:  (df['col'] - df['col'].mean()) / df['col'].std()
```

📈 Visualization & File Export

Built-in Plots

```
df['col'].hist(bins=20)
df.boxplot(column='num', by='cat')
df.plot.scatter(x='col1', y='col2')
```

Seaborn Integration

```
import seaborn as sns
sns.heatmap(df.corr(), annot=True)
sns.pairplot(df)
```

Export Data

```
df.to_csv('output.csv', index=False)
df.to_excel('output.xlsx')
df.to_json('output.json')
df.to_sql('table', connection)
```

Memory Optimization

```
df.memory_usage(deep=True)
df['col'].astype('category')  # save RAM
pd.read_csv('f.csv', chunksize=1000)
```

💡 Use sns.heatmap(df.corr()) to instantly spot highly correlated features before modeling.

🚀 Save this for your next Data Analysis project!

🔍

Always inspect first
df.info() + df.describe()

🧼

Clean before you analyse
Handle nulls & duplicates

📊

GroupBy is your best friend
for pattern discovery

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